

D.15 Extended Related Work for Theoretical Attention Problems

The static attention computation is asking the following question that given $Q, K, V \in \mathbb{R}^{n \times d}$, the goal is to $D^{-1} \exp(QK^\top)V$ where $D = \text{diag}(\exp(QK^\top)\mathbf{1}_n)$. [98] studied the static attention computation from both algorithm and hardness. On the positive, they provide an almost linear time algorithm to approximately compute the attention matrix. On the negative side, assuming a strong exponential time hypothesis (SETH), they prove a hardness result. Their hardness result is, unless SETH fails, there is no algorithm that runs in truly subquadratic time to approximately compute the attention matrix. Further, [100] considers the dynamic of attention computation problem. They also provide both algorithmic results and hardness results. In the work of [103], they consider the sparsification of attention matrix construction. In particular, they assume that situation that $d \gg n$, and show how to sparsify the columns of matrix Q . [103] provides two algorithms, one is a randomized algorithm, and the other is a deterministic algorithm. Differential privacy is a famous and textbook topic in graduate school, recently the work of [101] shows how to give a differentially private algorithm for computing the attention matrix. For a given $A \in \mathbb{R}^{n \times d}$ and vector $b \in \mathbb{R}^n$, [104] formulates and studies exponential regression $\min_x \|\exp(Ax) - b\|_2$. Then [105] considers the normalization factor in exponential regression and defines the softmax regression problem $\min_x \|(\exp(Ax), \mathbf{1}_n)^{-1} \exp(Ax) - b\|_2$. [107] moves the scaling factor from $\exp(Ax)$ to b and defines a rescaled softmax regression problem $\min_x \|\exp(Ax) - \langle \exp(Ax), \mathbf{1}_n \rangle \cdot b\|_2$.

D.16 Sparsity Preserving

Recall that in Figure 2, we observe that even when trained densely, the attention matrices of LLMs are over 95% sparse at inference time. Only 5% of the KV cache is sufficient for decoding the same output token at each generation step. Here, we provide some formal formulations for sparsity.

Definition D.33. Suppose the following conditions

- Let $S_0 \subset [m]$.
- Let $k = |S_0|$.
- Let $\tau \in (0, 1)$ denote a threshold for truncating the value.
- Let $\alpha \in (0, 1)$ denote a fraction of mass (larger than τ) outside S_0 .
- Let mapping $\mathcal{D} : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}^m$.
- For each $x \in \mathbb{R}^d$, $\mathcal{D}(x) \in \mathbb{R}^m$ is a vector that has length m .

We say the distribution $\mathcal{D}$ is (α, τ, k) -good if the following conditions hold

- For all $x \in \mathbb{R}^d$, $S_0 \subset \text{supp}_\tau(\mathcal{D}(x))$
- For all $x \in \mathbb{R}^d$, $|\text{supp}_\tau(\mathcal{D}(x)) \setminus S_0| \leq \alpha \cdot k$

Claim D.34. Suppose we sample n points $\{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^d$ from (α, τ, k) -good distribution uniformly at random, then we have

- $S_0 \subseteq \cap_{i \in [n]} \text{supp}_\tau(x_i)$
- $|(\cup_{i \in [n]} \text{supp}_\tau(\mathcal{D}(x_i))) \setminus S_0| \leq \alpha kn$

Proof. Since for all $i \in [n]$, we have $S_0 \subseteq \text{supp}_\tau(\mathcal{D}(x_i))$, thus

$$S_0 \subseteq \cap_{i \in [n]} \text{supp}_\tau(x_i).$$

Therefore we proved the first property.

We know that for all $i \in [n]$, we have $|\text{supp}_\tau(\mathcal{D}(x_i)) \setminus S_0| \leq \alpha kn$. Thus

$$|(\cup_{i \in [n]} \text{supp}_\tau(\mathcal{D}(x_i))) \setminus S_0| \leq \sum_{i=1}^n |\text{supp}_\tau(\mathcal{D}(x_i)) \setminus S_0| \leq n \cdot \alpha k$$

Therefore, we finish the proof for the second property. $\square$

D.17 Definition of Loss Function

In this section, we follow the theoretical softmax regression literature [105] and define a number of functions to make the calculations of gradient and Hessian convenient. We also proposed a new penalty term (ℓ_1 type sparsity penalty, see Definition D.41) into the final loss function, which is not studied in previous work [104, 105, 107, 110, 132, 133]. We first provide some function definitions.

Definition D.35 (Function u , [105]). *Given matrix $A \in \mathbb{R}^{n \times d}$, let function $u : \mathbb{R}^d \rightarrow \mathbb{R}^n$ be defined as follows*

$$u(x) := \exp(Ax)$$

Definition D.36 (Function α , see Definition 5.4 in [105] as an example). *We define $u(x)$ as Definition D.35. Then we define $\alpha : \mathbb{R}^d \rightarrow \mathbb{R}$ as follows*

$$\alpha(x) := \langle u(x), \mathbf{1}_n \rangle$$

Definition D.37 (Function f , see Definition 5.1 in [105] as an example). *Provided that the following conditions are true*

- We define $u(x)$ as Definition D.35.
- We define $\alpha(x)$ as Definition D.36

Let function $f : \mathbb{R}^d \rightarrow \mathbb{R}^n$ be defined as follows

$$f(x) := \alpha(x)^{-1}u(x).$$

Definition D.38 (Function c , see Definition 5.5 in [105] as an example). *Provided that the following conditions are true*

- Given a vector $b \in \mathbb{R}^n$.
- Let $f(x)$ be defined as Definition D.38.

Then, let function $c : \mathbb{R}^d \rightarrow \mathbb{R}^n$ defined as follows

- $c(x) := f(x) - b$.

Definition D.39 (Loss function L_{exp} , see Definition 5.3 in [105] as an example). *We define $L_{\text{exp}} : \mathbb{R}^d \rightarrow \mathbb{R}$*

$$L_{\text{exp}}(x) := 0.5 \cdot \|c(x)\|_2^2.$$

Definition D.40 (Loss function L_{reg}). *Given $A \in \mathbb{R}^{n \times d}$.*

Let function $L_{\text{reg}} : \mathbb{R}^d \rightarrow \mathbb{R}$ be defined as follows

$$L_{\text{reg}}(x) := 0.5 \cdot \|\text{diag}(w)Ax\|_2^2$$

We define a novel penalty function

Definition D.41 (Implicitly controlling the sparsity). *Given $A \in \mathbb{R}^{n \times d}$.*

We define

$$L_{\text{sparse}}(x) := \|\exp(Ax)\|_1.$$

Then it is obvious that we have

Claim D.42. *Given $A \in \mathbb{R}^{n \times d}$. Let u be defined as Definition D.35. Let α be defined as Definition D.36.*

We have

- $L_{\text{sparse}}(x) = \langle \exp(Ax), \mathbf{1}_n \rangle$
- $L_{\text{sparse}}(x) = \langle u(x), \mathbf{1}_n \rangle$

- $L_{\text{sparse}}(x) = \alpha(x)$

Proof. The proof is trivially following from the definition of $u(x)$ (see Definition D.35) and $\alpha(x)$ (see Definition D.36). $\square$

The final loss function can be defined as follows. Intuitively, we can write attention $D^{-1} \exp(QK^\top)$ into n subproblems where each subproblem can be viewed as one softmax problem.

Definition D.43. *If the following conditions hold*

- We define L_{exp} as Definition D.39.
- We define L_{reg} as Definition D.40.
- We define L_{sparse} as Definition D.41.

Then we define L function

$$L(x) := L_{\text{exp}}(x) + L_{\text{sparse}}(x) + L_{\text{reg}}(x).$$

D.18 Gradient

Next, we show the gradient of L_{exp} .

Lemma D.44 (Gradient, Lemma 5.6 in [105]). *Provided that the following conditions are true*

- Given matrix $A \in \mathbb{R}^{n \times d}$ and a vector $b \in \mathbb{R}^n$.
- Let $A_{*,i} \in \mathbb{R}^n$ denote the i -th column of matrix A , for every $i \in [d]$.
- We define $\alpha(x)$ as Definition D.36.
- We define $f(x)$ as Definition D.37.
- We define $c(x)$ as Definition D.38.
- We define $L_{\text{exp}}(x)$ as Definition D.39.
- Let $\circ$ denote hadamard product.

For every $i \in [d]$, we have

- Part 1.

$$\frac{d \exp(Ax)}{dx_i} = \exp(Ax) \circ A_{*,i}$$

- Part 2.

$$\frac{d \langle \exp(Ax), \mathbf{1}_n \rangle}{dx_i} = \langle \exp(Ax), A_{*,i} \rangle$$

- Part 3.

$$\frac{d \alpha(x)^{-1}}{dx_i} = -\alpha(x)^{-1} \cdot \langle f(x), A_{*,i} \rangle$$

- Part 4.

$$\frac{df(x)}{dx_i} = \frac{dc(x)}{dx_i} = -\langle f(x), A_{*,i} \rangle \cdot f(x) + f(x) \circ A_{*,i}$$

- Part 5.

$$\frac{dL_{\text{exp}}(x)}{dx_i} = A_{*,i}^\top \cdot (f(x)(f(x) - b)^\top f(x) + \text{diag}(f(x))(f(x) - b))$$